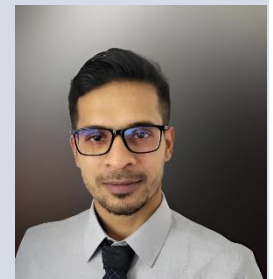


MD. SAIDUL ISLAM

Materials Engineer

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Profile Summary

Materials science professional, specialized in data-driven methods and characterization techniques. Experience with ML pipelines, FEM, materials informatics, and databases. Goal: Development of data-driven methods for materials discovery and process optimization.

Professional Experience

Timeline	Position	Task/achievement
09/2022 - 05/2023	Student Assistant, Multicomponent Materials, CAU, Kiel.	- Simulation of EM sensor (~2 mT). - Magnetostriction measured & analyzed (OriginPro). - Experiments designed with MATLAB. - AFM/TEM/profilometer characterization.
11/2021 - 12/2021	Student Assistant, CAU, Kiel.	Produced copper electrodes for implants and analyzed their light reaction with MATLAB.
03/2018 - 09/2019	Quality Engineer, Bandar Steel, Dhaka.	- Led production & quality control. - Reduced losses & alloy consumption by ~5%. - Analyzed breakdowns in CCM/rolling mills.
12/2016 - 01/2018	Quality Engineer, , Rahim Group,Dhaka.	- Tests with universal testing machine. - Introduced scrap metal management. - Executive Engineer in furnace installation (30 t).

Education

Timeline	Degree	Details
10/2019 - 03/2024	M.Sc., Materials Science & Engg., CAU, Kiel.	GPA: 2.2 / 5.0 (Best-GPA: 1.0) Master-thesis: Modeling of source nonlinearity in Electro-magnetic systems with SINDy & ANN.
02/2011 - 02/2017	B.Sc. Engg., Materials & Metallurgical Engg., BUET, Dhaka.	GPA: 2.94 / 4.0 (Best-GPA: 4.0) Bachelor-thesis: Reduction kinetics of mill scale analysis.
06/2008 – 07/2010	Higher Secondary (HSC), Stamford College, Dhaka. GPA: 2.2/5.0 (Best-GPA :1.0)	

Professional Development & Research Projects (Selected)

Timeline	Theme	Description
03/2024 - Heute	Materials Database (Flask/SQLite, Fly.io)	Proof of concept for a research database with automatic ETL pipeline and deployment. [Git , DEMO]
	Melting-point prediction	Two-level stacking (RF, XGBoost, LightGBM, MLP), $R^2 = 0.83$. [Git] (Citration, 3.041k samples).
	Oxidation-state Assignment	Soft-Voting-Ensemble (LGBM, RF,GB, ExtraTrees), $R^2 = 0.91$. [Git] (OQMD, ~7k samples).
	Semantic Modelling (LLM integrated)	OLLAMA (local)-coupled semantic knowledge graph for custom-featurized band gap data. [Git]
	Coursera-Certificates (Auswahl)	<i>Introduction to High-Throughput Materials Development</i> <i>Materials Data Science & Informatics</i> <i>Density Functional Theory</i>

Computer & Analytical Skills

Programming & Data Analysis:	Python (good); MATLAB, SQL; Basic: HTML, Fortran, C++, CSS, SPARQL, RegEx.
AI & ML:	Ensemble-Methods, PINN, SINDy, DNNS, GNNs, PyCaret, SHAP; Knowledge graph(RDF); Generative AI; Active Learning
Materials Informatics:	Matminer, Pymatgen, RDKit; OQMD, Materials Project, Citrination; Semantic Web; SchNet
Simulation & FEM:	COMSOL, Abaqus, SimScale.
Characterisation:	AFM, TEM, SEM, XRD, VSM, DSC/TGA.
Web & Databases:	Flask, Jinja, Docker, SQLite.

Conference

- Poster presentation, AIMSE 2023, Saarbrücken – SINDy + ANN for magnetoelectric sensors.

Languages

- Deutsch (Telc B1).
- Englis (fluent).
- Bengali (Native).

Engagement & Interests

- Organization of student programs, club activities, and sports tournaments.
- Sport (Cricket, Football, Badminton).
- Musik (Guitar), Traveling